

Accessible Slides in LaTeX

https://github.com/rhstanton/accessible_LaTeX

Version 1.5

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L^AT_EX and accessibility: new requirements

- From **April 24, 2026**, course materials must meet WCAG 2.1 Level AA accessibility standards
 - Applies even to password-protected course sites (e.g., **bCourses**).
- Many of us use **L^AT_EX** to create teaching materials, both slides and documents.
- **But standard L^AT_EX (especially Beamer) does not automatically generate accessible PDFs.**
- **Good news:** The L^AT_EX Project team has developed solutions.

What is this project?

- A practical template set for accessible $\text{\LaTeX}$ using the [\text{\LaTeX} Tagging Project](#)
 - **Two templates:** Articles (**article** class) and slides (**ltx-talk**)
 - **Shows:** What is shared and what differs
 - **Contains:** Math, text, figures, and tables
 - **Validation status:** Slides score 100% with [Ally](#) and pass common validators ([veraPDF](#), [PAC](#), [Acrobat](#)) in typical workflows (tested March 17, 2026).
- Also includes some (optional) additional slide features
 - Readable [outline slide](#)
 - Automatic [section separator slides](#)
 - Multi-page [bibliography support](#)
- Repo: https://github.com/rhstanton/accessible_LaTeX

Converting to accessible $\text{\LaTeX}$: the big picture

- **Your existing $\text{\LaTeX}$ skills transfer:** mainly adding a few consistent accessibility features.
- **Most changes are local:** metadata once; then per-figure and per-table tweaks.
- **Applies to both slides & articles:**
 - Document metadata + PDF tagging
 - Alt text on images
 - Table header row tagging
 - Sufficient color contrast
 - Compile with Lua $\text{\LaTeX}$
- **Key difference:**
 - Articles: **keep your existing class** and add tagging features
 - Slides: switch to `ltx-talk` (frame syntax similar to Beamer), then recreate styling

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1. Setup

Version requirements: what actually matters

- **Engine:** must compile with [Lua \$\LaTeX\$](#) (not pdf $\LaTeX$ or Xe $\LaTeX$).
- **LaTeX kernel/format date:** must be [2025-11-01 or later](#).
- **Important:** a “TeX Live year” alone is not a reliable guarantee.
 - You need a sufficiently new $\LaTeX$ kernel *on your install*, or use Overleaf Labs.
- **Quick check (in any document preamble):**

```
\show\fmtversion
```

Overleaf (works when configured correctly)

- `ltx-talk` requires a very recent $\text{\LaTeX}$ kernel (format date `2025-11-01+`).
- On Overleaf, this typically means using the `Labs` program (**not standard Overleaf**).
- **Overleaf setup (2 steps):**
 1. Join Overleaf Labs:
 - <https://www.overleaf.com/labs/participate>
 - Opt in and enable `Rolling TeX Live releases`
 2. Configure the project:
 - TeX Live version: `Rolling TeXLive (labs)` (bottom of list)
 - Compiler: `Lua $\text{\LaTeX}$`
- Overleaf guide: <https://docs.overleaf.com/writing-and-editing/creating-accessible-pdfs>

2. Core requirements

Core requirement 1: document metadata

- Required for both slides and articles
- Every accessible document **must** begin with \DocumentMetadata
- Goes **before** \documentclass

```
\DocumentMetadata{
  % Request both PDF/A archival conformance and PDF/UA accessibility tagging
  pdfstandard=a-2u,      % PDF/A-2u format
  pdfstandard=ua-1,      % Add PDF/UA-1 conformance target
  pdfversion=1.7,        % Explicit PDF version
  lang=en-US,            % Language for screen readers
  tagging=on,            % Enable PDF tagging
  tagging-setup={
    math/alt/use,         % Keep math accessibility enabled
    math/mathml/AF=false,% Do not embed MathML as PDF Associated Files
    math/tex/AF=false,    % Do not embed TeX source as Associated Files
    math/mathml/sources=  % Clear MathML source attachment list
  }
}
```

- **Why PDF 1.7 /UA-1?** Current checkers (PAC, Acrobat, Ally, veraPDF) are most reliable for PDF 1.7/UA-1 and may incorrectly flag valid PDF 2.0/UA-2 features
 - See [PDF Days 2025 \(LaTeX Tagged PDF talk\)](#)

Core requirements 2–4: images, tables, and color contrast

- **Required for both slides and articles.**
- **Images:** every `\includegraphics` needs `alt={...}`.
- **Tables:** mark header rows using `\tagpdfsetup{table/header-rows={...}}`.
- **Colors:** WCAG 2.1 AA requires [4.5:1](#) contrast ratio for normal text.
- **Accessible color defaults:**

```
\colorlet{AccessibleRed}{red!80!black}  
\colorlet{AccessibleGreen}{green!40!black}  
% Standard blue is fine
```

Core requirement 5: compile with Lua $\text{\LaTeX}$

- **Required for both slides and articles.**
- Compile with **Lua $\text{\LaTeX}$** (not pdf $\text{\LaTeX}$ or Xe $\text{\LaTeX}$).
- **Why Lua $\text{\LaTeX}$?**
 1. Automatic math accessibility support (no manual MathML files in normal workflows)
 2. Full tagging support
 3. Good Unicode/OpenType font support
- **How to switch:**
 - Command line: `lualatex filename.tex`
 - Editors: select LuaLaTeX as the compiler

3. Examples

Slide content: mostly the same as Beamer

- Slides live in `frame` environments (similar to Beamer).
- So your content often transfers with minimal changes.

- **Typical frame skeleton:**

```
\begin{frame}{Title}  
  \begin{itemize}  
    \item First point  
  \end{itemize}  
\end{frame}
```

- **Note (optional):** template fonts keep digits/percent/dollar signs consistent in math/text.
 - **Text:** \$1234567890%.
 - **Math:** \$1234567890%.

Figures: alt text

- Figures are included as usual (e.g., `\includegraphics`), but **add alt text**.

- Good (minimal):** `\includegraphics[alt={A capybara}]{capybara.jpg}`

- Better (more informative):**

```
\includegraphics[alt={A capybara's head facing left}]{capybara.jpg}
```

- Decorative images: use empty alt text: `alt={}`.



Tables: header rows

- Tables use tabular as usual, but you must specify **header rows**.
- Use {1} for one header row, {1,2} for two header rows, etc.
- **Note:** PAC may still warn about header associations in some cases; see [Validator-specific notes](#).

```
\tagpdfsetup{table/header-rows={1,2,3}}  
\begin{tabular}{cccrcccr}  
\toprule  
← 3 header rows  
\midrule  
← data rows  
\bottomrule  
\end{tabular}
```

Payment date	Caplet expiry date	DF_{pay}	Forward rate	Days to expiry	Days in accrual period	T_{expiry}	Δ	Caplet
2004/12/01	—	0.99550	0.01790	0	91	0.00000	0.25278	—
2005/03/01	2004/11/29	0.99008	0.02188	89	90	0.24384	0.25000	1,178.77
2005/06/01	2005/02/25	0.98401	0.02413	177	92	0.48493	0.25556	4,844.73
2005/09/01	2005/05/27	0.97733	0.02675	268	92	0.73425	0.25556	10,016.71

4. Slides versus articles

Articles vs. slides

- **Articles:** keep your existing class (e.g., `article`) and add the checklist items.
- **Slides:** switch to `ltx-talk`.
 - Your `\begin{frame}` content often stays the same.
 - Remove Beamer themes/templates; recreate styling with standard $\text{\LaTeX}$ packages (as in this template).
- **If you're converting Beamer slides:**

<code>\documentclass{ltx-talk}</code> (instead of beamer)

Getting started (new vs. converting): keep it simple

- **New documents:**
 - copy `accessible_slides.tex` or `accessible_article.tex`
 - replace content; keep the metadata/preamble structure
- **Converting existing documents:**
 - add metadata + Lua^AT_EX + alt text + table header rows
 - for slides, switch to `ltx-talk` and redo styling once
- Repo: https://github.com/rhstanton/accessible_LaTeX

5. Validation and known issues

Common pitfalls

- **Confusing PDF tagging with PDF bookmarks**
 - tagging=on creates the structure tree for **screen readers** (required).
 - PDF bookmarks (navigation pane) are separate and optional.
- **Forgetting alt text**
 - Every `\includegraphics` needs `alt={...}` (or `alt={}` if decorative).
- **Not tagging table header rows**
 - Add `\tagpdfsetup{table/header-rows={...}}` before each table.
- **Low contrast colors**
 - Avoid light colors (e.g., plain yellow/cyan).
 - Use darker variants for red/green (defined in this template).
- **Wrong engine or old kernel**
 - Use Lua $\text{\LaTeX}$ and ensure format date 2025-11-01 or later.

Validation workflow: use multiple checkers

- No single checker catches everything.
- Accessibility usability checks and strict PDF conformance checks are not identical.
- Practical workflow: run 2+ tools and compare findings.
- **Common tools:**
 - Ally (bCourses)
 - veraPDF (PDF/A conformance): <https://verapdf.org/>
 - PAC (PDF accessibility): <https://pac.pdf-accessibility.org/>
 - Adobe Acrobat (useful but has known quirks)

Optional cleanup for strict validators: strip_af.py

- The metadata settings alone can be enough for a 100% score in Ally.
- Some strict tools (e.g., veraPDF) may still complain about /AF or embedded-file structures.
- To remove these, run:

```
python strip_af.py build/accessable_slides.pdf
```

- Prerequisite: install the Python package pikepdf first: `python -m pip install pikepdf`.
- Then validate again with veraPDF.

Validator-specific notes (common false positives)

- **Acrobat: “Lbl and LBody – Failed” (validator bug)**
 - Often appears around figure captions, section numbers, table captions.
 - This is an Acrobat bug in common setups; typically safe to ignore if source tagging is correct.
 - Reference: <https://tex.stackexchange.com/a/709655/2388>
- **PAC table warning (known behavior)**
 - PAC can report: Table header cell has no associated subcells even when table/header-rows is correct.
 - Documented by the Tagging Project:
<https://github.com/latex3/tagging-project/issues/1056>
- **Bottom line:** treat one-tool warnings as investigation prompts, not definitive errors.

6. Quick start

Quick start: accessible $\text{\LaTeX}$ on one slide

- **To make an accessible $\text{\LaTeX}$ PDF:**
 1. Add `\DocumentMetadata{...}` (before `\documentclass`)
 2. Compile with $\text{\Lua\LaTeX}$
 3. Every figure: `\includegraphics[alt={...}]{...}`
 4. Every table: `\tagpdfsetup{table/header-rows={...}}`
 5. Use high-contrast colors (4.5:1) and validate with 2+ tools
- Template repo: https://github.com/rhstanton/accessible_LaTeX
- Questions: richard.stanton@berkeley.edu

7. (Optional) additional utilities

Optional presentation utilities

- This template includes **optional utilities** in `slide_utils.sty`
 - **Not required for accessibility**
 - Useful for common presentation tasks
 - Easy to customize or remove
- **Two optional features:**
 1. Automatic section separator slides
 2. Multi-page bibliography support
- See `slide_utils.sty` for implementation details and customization options.
- **Note:** There's additional code in the main preamble to get readable table of contents.

Optional feature 1: section separator slides

- **What it does:** Automatically inserts a separator slide after each `\section` command
 - Displays section number and title in large text
 - Centered on the slide for visual emphasis
 - Uses accessible colors (AccessibleRed for sections, blue for subsections)
- **Example:** After `\section{Core requirements}`, automatically creates a slide saying

3. Core requirements

- **To disable:** Comment out or remove the redefinition code in `slide_utils.sty` Section 1
- **To customize:** Edit colors, fonts, or spacing in the frame template

Optional feature 2: multi-page bibliography

- **Problem:** `\bibliography` inside frame environment puts all references on one slide.
 - For large bibliographies, this can be unreadable.
 - This was easy with Beamer, using the `allowframebreaks` option.
 - But `ltx-talk` does not have a similar option.
- **Solution:** Write code to split bibliography automatically across multiple slides
 - Works with standard BibTeX workflow
 - Uses Lua script ([rsbibsplitsplit.lua](#)) to split the `.bbl` file
 - (Lua^ATeX provides access to the [Lua](#) programming language)
- **Usage:** As usual (see end of this `.tex` file for complete example)
 - `\cite{key1,key2,...}` to specify references
 - `\bibliography{bibfile}` to specify bib file
 - (Set `\RS@bibmax` to customize max per slide; default=8).
 - **Note:** Do not put `\bibliography` command inside frame environment
- See `slide_utils.sty` Section 2 for detailed workflow explanation.

8. References

References

- Akerlof, G. A., 1970, The market for “lemons”: Quality uncertainty and the market mechanism, *Quarterly Journal of Economics* 84, 488–500.
- Arrow, K. J., 1963, Uncertainty and the welfare economics of medical care, *American Economic Review* 53, 941–973.
- Barro, R. J., 1991, Economic growth in a cross section of countries, *Quarterly Journal of Economics* 106, 407–443.
- Black, F., and M. Scholes, 1973, The pricing of options and corporate liabilities, *Journal of Political Economy* 81, 637–654.
- Coase, R. H., 1960, The problem of social cost, *Journal of Law and Economics* 3, 1–44.
- Fama, E. F., 1970, Efficient capital markets: A review of theory and empirical work, *Journal of Finance* 25, 383–417.
- Fama, E. F., and K. R. French, 1993, Common risk factors in the returns on stocks and bonds, *Journal of Financial Economics* 33, 3–56.
- Jensen, M. C., and W. H. Meckling, 1976, Theory of the firm: Managerial behavior, agency costs and ownership structure, *Journal of Financial Economics* 3, 305–360.

References

- Kahneman, D., and A. Tversky, 1979, Prospect theory: An analysis of decision under risk, *Econometrica* 47, 263–292.
- Krugman, P. R., 1991, Increasing returns and economic geography, *Journal of Political Economy* 99, 483–499.
- Lucas, R. E., Jr., 1976, Econometric policy evaluation: A critique, *Carnegie-Rochester Conference Series on Public Policy* 1, 19–46.
- Merton, R. C., 1974, On the pricing of corporate debt: The risk structure of interest rates, *Journal of Finance* 29, 449–470.
- Modigliani, F., and M. H. Miller, 1958, The cost of capital, corporation finance and the theory of investment, *American Economic Review* 48, 261–297.
- Rothschild, M., and J. E. Stiglitz, 1976, Equilibrium in competitive insurance markets: An essay on the economics of imperfect information, *Quarterly Journal of Economics* 90, 629–649.
- Solow, R. M., 1956, A contribution to the theory of economic growth, *Quarterly Journal of Economics* 70, 65–94.